

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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ABSTRACT

Annotation scarcity is a pervasive obstacle to training robust vision models across the many specialized domains—medical imaging prominent among them—where expert labels are costly and hard to obtain. Self-supervised masked image modeling (MIM) is a broadly promising remedy, yet mainstream approaches lean on random high-ratio masking, which tends to be inefficient and semantically poorly aligned across the range of downstream objectives. Region-aware variants only partially close this gap: because they typically hinge on reconstruction heuristics or task-specific supervised signals, their adaptability across tasks, modalities, and deployment settings remains limited. We propose Mask What Matters, a controllable text-guided masking framework for self-supervised image analysis that we study in the medical domain. By coupling vision-language models with prompt-based region localization, the method flexibly steers differentiated masking toward the semantically salient regions that matter for a given task while suppressing redundancy in background areas. Because guidance is expressed through open-vocabulary language rather than fixed heuristics, this controllable design promotes stronger semantic alignment, more transferable representation learning, and generalization that carries across diverse tasks. Evaluated across a broad span of medical imaging modalities—including brain MRI, chest CT, and lung X-ray—Mask What Matters consistently outperforms representative MIM methods (e.g., SparK), with gains reaching +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection. Notably, these improvements hold even when the overall masking budget is markedly reduced (e.g., 40% vs. 70%), pointing to efficiency benefits that generalize across settings. Taken together, these results indicate that controllable, text-driven masking is a broadly applicable route to semantically aligned self-supervised learning, and one that can advance robust vision models well beyond any single imaging modality.

1 INTRODUCTION

Image classification, segmentation, and object detection are fundamental tasks that span virtually the whole of computer vision. In recent years, machine learning—and deep learning in particular—has become the core technology driving progress across these tasks and their many application areas. Yet high-performing deep learning models generally depend on large-scale annotated datasets, and this dependence becomes a decisive bottleneck wherever expert labels are scarce or expensive. Specialized domains such as medical image processing exemplify this challenge especially acutely: high-quality annotations are difficult and costly to obtain, posing a significant barrier to the broad adoption of deep learning methods in these settings.

Self-supervised learning (SSL) offers a broadly applicable solution by exploiting abundant unlabeled data for pretraining, enabling models to learn effective feature representations without manual annotations [Jaiswal et al. \(2020\)](#); [Liu et al. \(2021\)](#); [Krishnan et al. \(2022\)](#); [Huang et al. \(2023\)](#). Among the various SSL paradigms, MIM has emerged as a particularly widely adopted approach. By masking parts of an image at random and requiring the model to reconstruct the original content, MIM encourages the model to capture structural and semantic patterns from the visible context. This

objective has proven effective across a wide range of downstream tasks and data domains, and it has transferred naturally to medical image analysis as well [He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)].

However, prevailing MIM approaches—such as masked autoencoders (MAE) [He et al. (2022)] and SparK [Tian et al. (2023)]—generally adopt random masking strategies that are not aligned with the regions of interest of any particular downstream task. This task-agnostic masking gives rise to two limitations that recur across tasks and domains: (1) The learned representations can be semantically misaligned with the downstream objectives, which erodes transfer performance. (2) To raise the chance of masking informative areas, existing methods tend to rely on extremely high masking ratios (e.g., 75% in MAE), which increases reconstruction difficulty and, in turn, demands large-scale data and substantial computational resources. Natural language, by contrast, is compact and semantically dense, which enables effective learning even at low masking ratios [Devlin et al. (2019)]. The reliance on high masking ratios in image-based MIM stems largely from the inherent information redundancy of images and the absence of semantic-aware masking, which together make it difficult for models to concentrate on the regions that truly matter for the task at hand.

This issue is thrown into especially sharp relief by medical imaging, which serves as a demanding and representative testbed for these challenges. Medical images exhibit pronounced semantic sparsity: diagnostically relevant information is frequently confined to small localized regions (e.g., lesions or organs), while a large fraction of the image consists of semantically redundant background. Medical imaging further spans a broad set of modalities—including MRI, CT, and X-ray—with substantial variation in the shape, size, and spatial distribution of task-relevant areas from one task to the next. Such heterogeneity places stringent demands on the efficiency and cross-setting generalizability of masking strategies, and any strategy that meets them here is well positioned to transfer to other domains with similar structure.

To address these challenges, we introduce a controllable, text-guided masking framework—hereafter referred to as **Mask-What-Matters (MWM)**. Given any user-specified, open-vocabulary description or phrase, a pretrained vision–language model supplies zero-shot localization cues that we convert into robust regions of interest (ROIs). MWM then applies region-specific, prompt-conditioned masking—assigning higher ratios to semantically important areas (e.g., lesions or organs) and lower ratios to background—thereby injecting downstream semantics directly into the pretraining signal without per-image reports or labels. Because the guidance is driven by open-vocabulary language rather than fixed, domain-specific rules, the same mechanism adapts to new tasks and modalities simply by changing the prompt.

To summarize, our main contributions are:

- The first controllable text-guided masking framework for medical imaging, and one whose design principles extend naturally to other data domains. MWM integrates vision–language models (BiomedCLIP) with segmentation refinement (SAM) to localize task-relevant regions from open-vocabulary prompts and apply differentiated ROI vs. background masking, overcoming the semantic misalignment inherent to random masking.
- An annotation-free and backbone-agnostic design. MWM requires no per-image reports or labels and can be seamlessly integrated into both ViT- and ConvNet-based MIM pipelines under a single unified protocol, making it readily portable across architectures and pipelines.
- Consistent gains across modalities and tasks. Across brain MRI, chest CT, and X-ray datasets, MWM surpasses state-of-the-art MIM baselines (e.g., SparK), improving classification, detection, and segmentation alike—evidence that text-driven generative pretraining is broadly effective in medical imaging and a promising direction for vision–language pretraining more generally.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling first arose in natural language processing with the bidirectional encoder representations from transformers (BERT) model [Devlin et al. (2019)], which learns contextual representations by masking tokens and predicting them during pretraining. Inspired by this idea, [He et al. (2022)] carried MIM over to computer vision, where portions of an image are masked during pre-

training and the model is tasked with reconstructing the missing regions. The paradigm has since shown strong generalization across both natural and medical image domains (Gupta et al. 2025; Liu et al. 2021; Huang et al. 2023). Representative methods span a range of architectures, including MAE (He et al. 2022), MedMAE (Gupta et al. 2025; Xiao et al. 2023), and SparK (Tian et al. 2023). Among them, MAE uses high-ratio random masking to encourage ViTs to focus on high-level semantic representations; MedMAE carries this idea into medical imaging through domain-specific adaptations; and SparK ports the strategy to convolutional backbones, exploiting their inherent multi-scale features to strengthen pretraining. Together they illustrate the versatility of MIM across both transformer and convolutional families.

Despite these advances, conventional MIM approaches generally adopt random or structure-agnostic masking that ignores the semantic requirements of whatever downstream task is targeted. Such non-selective masking can lead the model to overemphasize irrelevant regions during pretraining, diminishing transfer effectiveness across tasks. To close this gap, a growing body of work has explored region-aware masking strategies aimed at improving semantic alignment, and two broad directions have emerged:

One line of work introduces loss-driven adaptive masking, in which methods such as hard patches mining (HPM) (Wang et al. 2023a) and its extension AnatoMask (Li et al. 2024) dynamically analyze reconstruction errors across regions and assign higher masking probabilities to areas with larger losses, seeking to prioritize semantically valuable structures. This heuristic, however, can steer the model toward regions that are merely hard to reconstruct rather than genuinely task-relevant. In brain magnetic resonance imaging (MRI), for instance, complex structures such as the spinal cord or eye-balls may draw unnecessary attention even though they bear little relation to downstream objectives such as tumor detection—an alignment problem that recurs whenever reconstruction difficulty is used as a proxy for task relevance.

A second line of work brings in external perceptual modules to guide masking. FocusMAE (Basu et al. 2024), for example, leverages pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. Although effective on the tasks it targets, this strategy requires supervised training of RPNs on domain-specific annotated data, which constrains its portability to new tasks and settings.

These limitations point to a shared need for a masking mechanism that is at once more flexible, more semantically aligned, and more broadly generalizable. Natural language—a rich, interpretable, and open-ended source of supervision—offers a compelling way to guide masked modeling in a task-aware yet annotation-free manner that is not tied to any single task or modality.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

The rapid maturation of vision-language alignment models has made it increasingly feasible to perceive and identify task-relevant regions accurately under zero-shot and open-world conditions (Radford et al. 2021). Foundational approaches such as the Contrastive Language-Image Pretraining (CLIP) family learn a unified embedding space for images and text through large-scale joint pretraining on image-text pairs, and have been adopted across a wide array of zero-shot and open-set visual tasks. Within medical imaging, extensions such as BiomedCLIP (Zhang et al. 2023) further strengthen cross-modal understanding and deliver robust performance across a diverse set of downstream applications. Yet even though vision-language pretrained models are now widely used across many supervised learning tasks, most existing studies confine them to downstream applications such as object detection and semantic segmentation (Zhong et al. 2022; Aleem et al. 2024; Koleilat et al. 2024), leaving their open-vocabulary, zero-shot capabilities largely untapped as a source of guidance for masked image modeling.

To summarize, although recent region-aware MIM approaches—whether based on dynamic reconstruction-loss masking or on masking guided by external perception modules—have sharpened semantic focus to a degree, they continue to share several limitations: a reliance on additional supervision, a persistent mismatch between masked regions and task semantics, and limited flexibility when moving across tasks and modalities. At the same time, despite the strong downstream performance of vision-language models, little effort has gone into folding natural-language prompts into the MIM pretraining phase to guide task-aware masking. Read together, these observations point

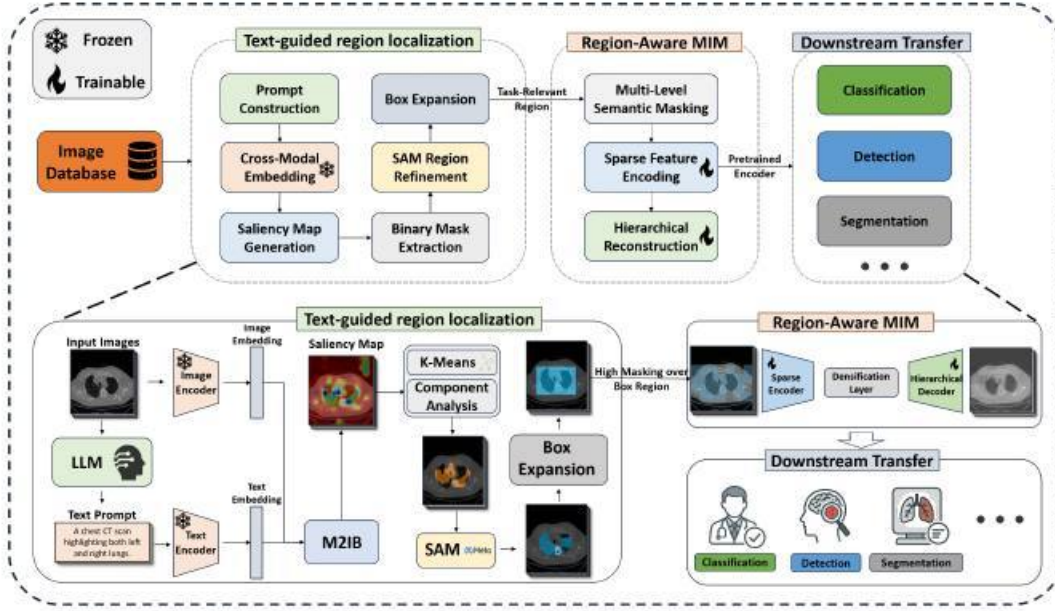


Figure 1: Overview of the MWM framework, presented as a general text-guided masking pipeline instantiated here for medical imaging. The top panel traces the three stages: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel zooms in to show how open-vocabulary prompts steer region localization and the resulting masking.

toward a promising and broadly applicable direction: using vision-language semantic guidance to make masked image modeling both more efficient and more generalizable.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** is organized around two core stages that together form a general, prompt-driven recipe: (1) text-guided region localization and (2) region-aware masked image modeling. The first stage uses a frozen vision-language model to identify task-relevant regions from an open-vocabulary prompt—so that the notion of “what matters” is set by language rather than fixed in advance—while the second stage applies differentiated masking ratios across those regions to guide reconstruction.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions through a modular multi-stage pipeline whose components can be swapped as needed for different settings. It begins with LLM-based prompt generation and employs BiomedCLIP to extract cross-modal embeddings. A saliency map is then produced via the Multi-Modal Information Bottleneck (M2IB), binarized with K-Means, and refined using the Segment Anything Model (SAM). Finally, the refined mask is converted into a bounding box with a controllable expansion margin, yielding a robust spatial guide for the subsequent masked modeling stage.

Prompt Generation. Across image-analysis settings, and medical imaging in particular, the design of textual prompts is a key determinant of how effective region localization will be. To strengthen semantic guidance, large language models (LLMs) are queried with structured templates such as “Describe the typical visual characteristics of a [category] in a [modality] image.” The [category] and [modality] slots make the template reusable across tasks and imaging types. Alongside these

full-sentence prompts, simple category phrases are also employed as alternative inputs, and a detailed comparison of the two prompt styles is presented in the experimental section.

Cross-Modal Embedding. Once the prompt is generated, a BiomedCLIP model [Zhang et al. (2023)] encodes the image and its prompt into a shared semantic space; any comparable image-text encoder could stand in for domains beyond the medical one. Given an image I and a prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These cross-modal embeddings supply the semantic foundation on which saliency estimation is built in the subsequent stages.

Saliency Map Generation. A cross-modal saliency estimation module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al. (2023b)] localizes task-relevant visual regions by aligning image and text embeddings while suppressing modality-specific redundancy—a formulation that applies whatever the prompt or imaging type. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns each spatial location an importance weight reflecting its semantic relevance to the prompt. M2IB obtains this map by optimizing the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are the image and text embeddings encoded by BiomedCLIP, and γ tunes the trade-off between preserving cross-modal relevance and filtering out task-irrelevant visual information—a single knob that generalizes across prompts and modalities.

Binary Mask Extraction. Given the saliency map, we binarize it with unsupervised K-Means clustering [Lloyd (1982)] to localize the prompt-guided region and produce a preliminary foreground mask. The map is clustered into two pixel groups, and the cluster with higher saliency values is taken as foreground. Because this step is fully unsupervised, it requires no threshold tuning per dataset. To refine the mask further and suppress spurious activations, connected component analysis is applied to the binary map, retaining only the largest foreground regions; this filters out noisy edge areas and concentrates attention on the core semantic structures.

Region Refinement. To improve spatial accuracy, SAM is incorporated as a refinement module [Kirillov et al. (2023)], chosen for its strong zero-shot segmentation across image types. For each selected connected region $c_i \in \mathcal{C}^*$, its minimal enclosing bounding box is computed and used as the visual prompt for SAM.

Given the original image I and the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

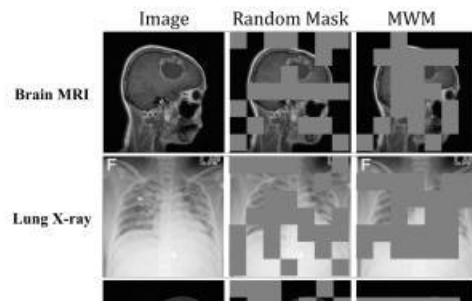
The resulting mask M_{SAM} provides more faithful spatial structure.

Box-Based Region Expansion. The SAM mask from the previous stage can carry boundary noise that would otherwise mislead the model’s focus during representation learning. To make region guidance more robust across varied region shapes and sizes, the refined SAM mask is converted into a bounding box with a controllable expansion margin, which then serves as the final region of interest (ROI) for downstream masked modeling.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the region of interest produced by the localization module in Section 3.2, MWM carries out masked image modeling through a three-stage pipeline that is agnostic to the specific task or modality: (1) a multi-level masking strategy that assigns differentiated masking ratios according to region importance, (2) a sparse encoder that processes only the unmasked patches, and (3) a hierarchical decoder that reconstructs the full image from sparse multi-scale features.

Multi-Level Semantic Masking. MWM adopts a multi-level masking strategy driven by the semantic regions surfaced during text-guided localization. Patches falling within regions highlighted by the prompts—for example, those associated with tumors



or organs, but in general whatever the prompt designates—are masked at a high ratio, pushing the model to reconstruct task-relevant semantic features. Background regions, by contrast, are masked lightly or left unmasked to curb redundant computation. Because the split between high- and low-ratio regions is dictated by the prompt rather than by fixed rules, the same scheme accommodates task-relevant regions of widely differing shape and extent. A visual comparison of the resulting masking patterns is shown in Figure 2.

Sparse Feature Encoding. Visible patches are collected into a sparse image and encoded following the Spark design (Tian et al., 2023):

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i is the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ are multi-scale features. Although we instantiate MWM with convolutional backbones here, the same interface extends directly to ViTs via token indexing, so the encoding stage is not tied to any one architecture family.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image in a top-down manner, a design that scales gracefully across input types. At each scale l , masked locations are filled with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , then upsampled and fused with features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

Finally, the model is trained with mean-squared error on masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ is the set of masked patch indices, x_i the original patch, and $\hat{x}_i$ its reconstruction.

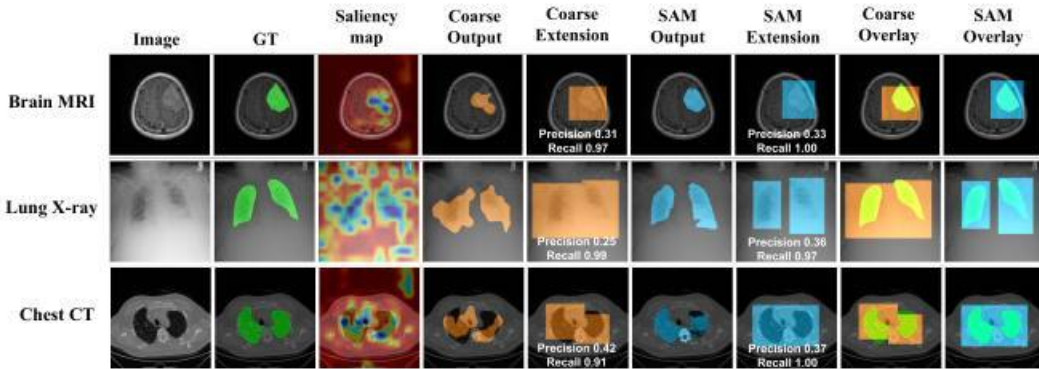


Figure 3: Representative text-guided region localization results spanning multiple imaging modalities.

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Localization performance (Occlusion Precision/Recall) reported across all three modalities under varying prompt types and refinement strategies. *Coarse Ext.* denotes bounding-box extension of the coarse region; + *SAM Ext.* indicates further refinement using SAM.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. To probe generalization across imaging modalities, self-supervised pretraining is conducted on three datasets that together span a broad range of anatomy and acquisition types. (1) *Brain MRI* comprises roughly 17,000 images drawn from Brain Tumor MRI [Cheng et al. (2015)], BRISC [Fateh et al. (2025)], and BraTS 2018 [Menze et al. (2014)], spanning diverse tumor types and anatomical structures. (2) *Lung X-ray* contains about 39,000 chest radiographs from COVID-QU-Ex [Rahman et al. (2021); Chowdhury et al. (2020)], encompassing COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* consists of 12,000 slices from a lung disease dataset [Konya (2020)], with segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. Downstream evaluation spans classification and detection and again covers all three modalities, so that transfer is measured under varied conditions rather than a single one. (1) *Brain Tumor MRI* [Feltrin (2023)] includes 4,479 MRI images spanning 44 tumor types, such as astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* [Kermany et al. (2018)] consists of 5,863 pediatric chest X-rays labeled as either normal or pneumonia. (3) *Chest CT Cancer* [Hany (2020)] contains CT images from four classes: adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue.

4.2 EXPERIMENTAL SETUP

To keep comparisons fair across the two architecture families, all self-supervised methods use backbones of comparable scale: convolution-based methods adopt ResNet-50 [He et al. (2016)] and transformer-based methods use ViT-S [Dosovitskiy et al. (2020)]. The input resolution is 224×224, and models are initialized from ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} , and downstream fine-tuning retains the same encoder and input resolution. All experiments run on NVIDIA Tesla A100 cards with 80 GB VRAM under Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise noted, non-target variables (e.g., prompt type or SAM usage) are held at their empirically optimal settings during evaluation, so that each comparison isolates the factor under study.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first assess the text-guided localization module across all three imaging modalities—brain MRI, lung X-ray, and chest CT—and then examine how prompt type and SAM refinement affect performance. Predicted bounding boxes are compared against expert-annotated segmentation masks, and we report occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), where R_p is the predicted region and R_{gt} is the ground-truth annotation.

As Table 1 shows, text-guided localization with effective prompts attains consistently high occlusion recall across every dataset (around 0.82) while keeping precision reasonable. This uniformity across modalities confirms that the framework can identify task-relevant regions from textual de-

scriptions alone, even as the underlying anatomy varies substantially from one modality to the next (see Figure 3 for visualized examples).

Effect of Prompt Type on Region Localization. To gauge how much language design matters, we compare two prompt types. As Table 1 indicates, descriptive sentence prompts work best on Brain MRI and Chest CT, suggesting that richer semantics aid localization where targets are compact. Concise phrases, however, win on lung radiographs in recall (0.82 vs. 0.66), likely because the lungs’ broad anatomical coverage lets general terms suffice. Rather than a single prompt style dominating everywhere, the results show that the framework accommodates a spectrum of prompt granularities, with the best choice adapting to the structure of the target region.

Effect of SAM Refinement on Region Localization. We also isolate the contribution of spatial refinement by applying SAM to the coarse masks. As Table 1 shows, SAM refinement raises precision while preserving high recall—on Brain MRI, precision climbs from 0.22 to 0.30 as recall stays above 0.94. The consistency of this pattern across modalities indicates that SAM is a broadly reliable way to sharpen coarse localization outputs.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We next probe the generalization and representation ability of MWM by comparing it against representative self-supervised learning (SSL) methods drawn from both the reconstruction-based and the contrastive paradigms, so that the comparison spans the major families of SSL rather than a single one. All methods are pretrained and evaluated under identical datasets and training settings to keep the comparison fair.

Image Classification. Classification performance on three downstream medical imaging datasets is examined. Following common practice, results are reported under two settings: (1) *full fine-tuning*, where the entire model is updated; and (2) *linear probing*, where the encoder is frozen and only a linear classifier is trained on top.

- **Full fine-tuning:** Table 2 summarizes classification accuracy for MWM alongside a broad slate of representative self-supervised methods, spanning MAE He et al. (2022), SparK Tian et al. (2023), AnatoMask Li et al. (2024), SimCLR Chen et al. (2020a), MoCoV2 Chen et al. (2020b), and BYOL Grill et al. (2020). MWM records the highest accuracy on every modality, evidence of generalization that holds across the full range of imaging types tested rather than favoring any one. The largest margin appears on Chest CT, where MWM reaches 97.5% accuracy—ahead of SparK by +3.1 and AnatoMask by +1.7. This gain underscores the payoff of text-guided masking, which concentrates on semantically critical yet spatially sparse regions—such as lung lobes in CT scans—while steering clear of redundant masking.
- **Linear probing:** To further probe representation quality, we run linear probing experiments. For a focused comparison, MWM is set against two recent state-of-the-art masked image modeling methods—SparK and AnatoMask. As Table 3 shows, MWM outperforms both on every dataset, confirming that the features learned through text-driven pretraining transfer well even without fine-tuning the encoder.

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Fine-tuning classification accuracy (%) across the three downstream modalities, where MIM denotes Masked Image Modeling and CL denotes Contrastive Learning.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear probing accuracy (%) across the downstream modalities with a frozen encoder.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) results (%) on the BR35H dataset, extending the evaluation from classification to localization- and pixel-level tasks.

Object Detection and Instance Segmentation. To test whether the learned representations generalize beyond classification to spatially demanding tasks, we evaluate brain tumor detection and instance segmentation on the BR35H dataset Hamada (2025). As Table 4 reports, MWM attains the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), exceeding SparK by +1.3 and +1.1 points, respectively. Together these results show that MWM transfers effectively to both object-level localization and pixel-wise segmentation, not just image-level recognition.

In sum, folding text-driven masking into pretraining yields representations that are both more informative and more transferable. The same semantic guidance lifts performance across classification, detection, and segmentation alike—breadth that points to text-guided masking as a general enhancement to visual representation learning rather than a task-specific trick.

4.5 ABLATION STUDY

We further conduct ablation experiments that isolate, across all three modalities, the individual impact of prompt design, SAM-based region refinement, and masking ratio on MWM’s downstream classification performance.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As Table 5 shows, descriptive sentence prompts outperform short phrases on brain MRI and chest CT, whereas the reverse holds on lung X-ray. This mirrors the localization results in Section 4.3: prompts that pinpoint regions more accurately also yield better downstream accuracy, and no single prompt style is universally optimal. Crucially, prompts of either granularity consistently beat random masking on every dataset. This uniform advantage confirms that text-guided masking establishes semantic alignment between pretraining and downstream tasks—alignment that proves broadly beneficial to self-supervised learning rather than dependent on a particular modality. The SAM ablation tells a similarly consistent story: adding the module during pretraining improves downstream classification across the board, confirming its value as a general enhancement to text-guided masking.

4.5.2 EFFECT OF MASKING RATIO

As Figure 4 shows, MWM reaches 96.8% classification accuracy at a masking ratio of just 40%—far below the roughly 70% commonly used in masked image modeling. Even at this reduced budget, it exceeds SparK’s best result at 70% masking by +2.4% in absolute accuracy. To rule out that SparK simply suffers from a

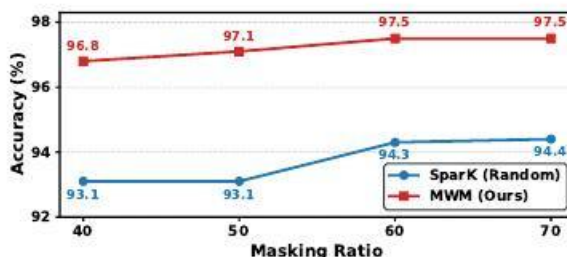


Figure 4: Classification performance on the Chest CT dataset across a range of masking strategies and ratios.

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under varying prompt types and region-refinement strategies, reported across all three imaging modalities. *No Prompt* denotes random masking without text-guided localization. *w/* and *w/o* indicate with and without SAM, respectively.

poorly chosen configuration, we sweep it over 40%, 50%, 60%, and 70% masking ratios; 70% indeed proves best for SparK, which validates the fairness of the comparison. These findings push back on the widely held assumption that high masking ratios are inherently optimal for masked image modeling. They instead point to a more general principle: semantically guided, task-aware masking such as MWM’s can deliver stronger downstream performance at lower masking ratios by preserving task-relevant content and discarding redundancy—an efficiency argument that should extend to masked modeling well beyond the specific setting studied here.

5 CONCLUSION AND LIMITATIONS

This work presents **Mask What Matters (MWM)**, a self-supervised pretraining framework that unites text-guided semantic localization with region-aware masking. By using natural-language prompts to highlight task-relevant areas and applying differentiated masking, MWM enables semantically aligned representation learning without per-image annotations—a recipe whose ingredients are, by design, not specific to any single task or modality.

Our experiments demonstrate the effectiveness and generalizability of MWM along several axes: (1) natural-language prompts reliably guide semantic region localization across modalities, establishing the feasibility of text-driven masking as a general mechanism; (2) MWM consistently outperforms existing methods (e.g., SparK, AnatoMask) in classification across all three imaging modalities, and delivers further gains in detection and instance segmentation; (3) MWM sustains strong performance even at lower masking ratios (e.g., 40%), highlighting the broad efficiency benefit of semantic guidance in self-supervised pretraining.

Limitations. Promising as these results are, several limitations remain and help delimit the scope of our claims. First, although text prompts provide flexible guidance, we have not systematically examined how robust MWM is to variation in prompt style, vocabulary, or noise. Second, the reliance on external textual descriptions introduces a dependency that could weaken generalization wherever reliable prompts are unavailable or inconsistent. Characterizing behavior under these conditions is a natural next step toward understanding how far the approach extends.

Looking ahead, we view text-driven, region-aware masking as a principled and broadly applicable route to more semantically grounded self-supervised learning. Because its core mechanism is domain-agnostic, extending MWM beyond medical imaging—to other modalities and to broader vision–language pretraining—stands out as a particularly promising avenue for future work.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

We used LLMs *only* for language editing and presentation polishing (e.g., grammar, phrasing, and minor stylistic clarity) of text written by the authors. LLMs were *not* used for idea generation, experimental design, data annotation, code implementation, or analysis/interpretation of results. All technical content, algorithms, proofs, figures, tables, metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.